Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 2

The Research Lifecycle

And first steps towards better data management

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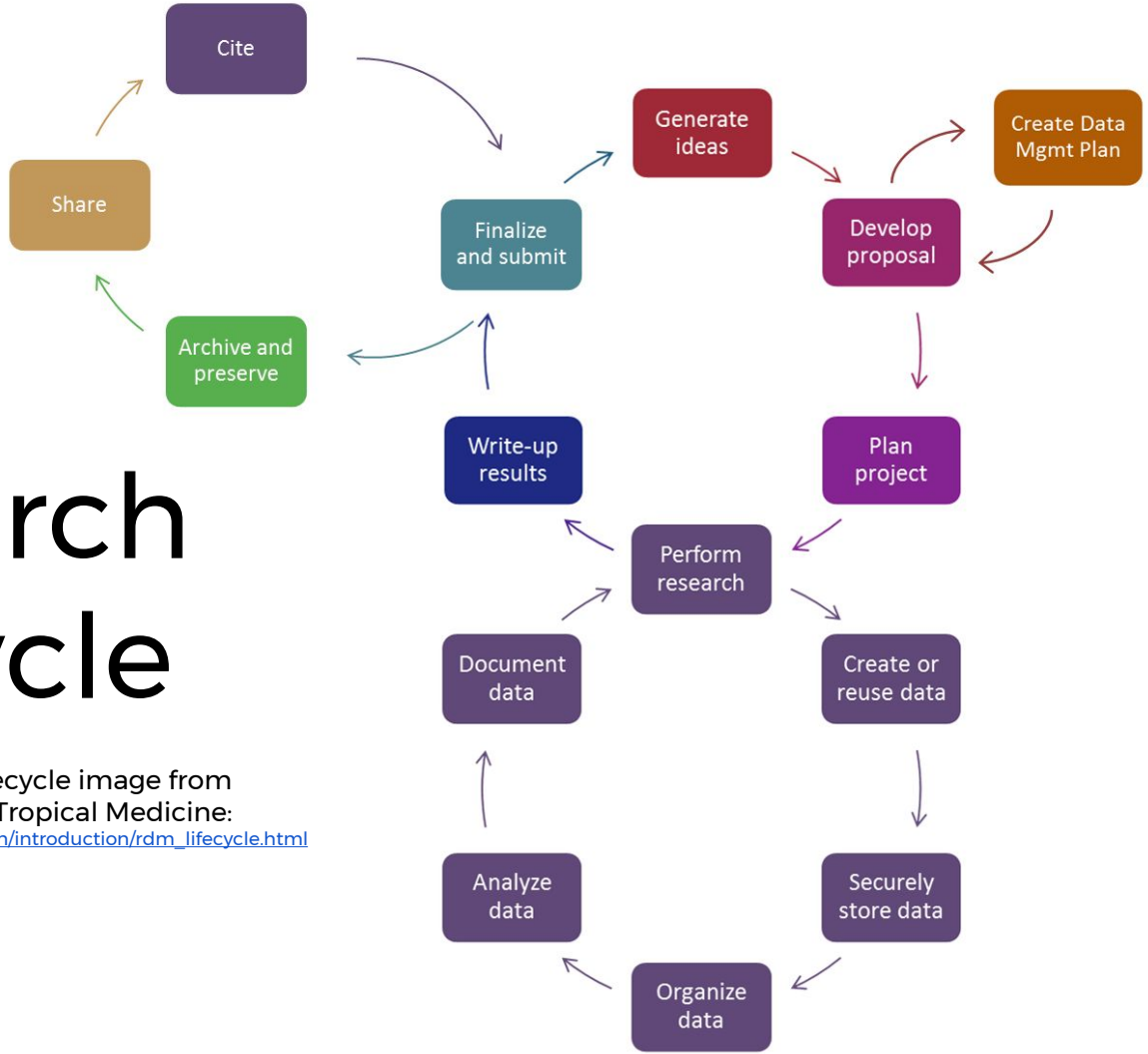
Learning Objectives

1. Describe the research lifecycle and the data lifecycle
2. Identify common data formats and file types that facilitate long-term access to data
3. Explain the significance of observing file naming conventions and version control
4. Explain the documentation needed to facilitate accessibility and reproducibility of findings by you or someone unfamiliar with your project

Research Lifecycle



Based off of the Research Lifecycle image from London School of Hygiene & Tropical Medicine:
http://www.lshtm.ac.uk/research/researchdataman/introduction/rdm_lifecycle.html

Preferred Research Data Formats

Data formats that offer the best chance for long-term access are **both**:

- Non-proprietary (also known as *open*), **and**
- Unencrypted and uncompressed

Examples of various data and file types that are better for long term preservation purposes include:

- Tabular Data: CSV
- Moving Images: MOV, MPEG
- Audio: WAVE, MP3
- Images: TIFF, JPEG 2000 (.jp2)
- Text: PDF/A, ASCII

Why Use File Naming Conventions?

Naming conventions make life easier!

- Help you find your data
- Help others find your data
- Help track which version of a file is most current

What File Naming Convention Should I Use?

Has your research group established a convention?

If not, general guidelines include:

- Meaningful file names that aren't too long
- Avoid special characters (* & % # . /)
- Dates can help with sorting and version control
  - ISO standard for dates: yyyymmdd

How does version control work?

Version control is a systematic approach to managing changes in documents.

- Changing document information (version numbers, dates)
- Systems (git, Open Science Framework)

Describing Data, Documenting Reliability & Collection Techniques

Who	Who collected this data? Who or what were the subjects under study?
What	What data was collected, and for what purpose? What is the content and structure of the data?
Where	Where was this data collected? What were the experimental conditions that produced it?
When	When was the data collected? Is the data part of a series, or ongoing experiment?
Why	Why was this experiment performed? How does it relate to your research question?

Why are documenting, versioning, and naming files so important?

<u>Consider this</u>:

If you unexpectedly have to leave your research project for a few months, could a colleague easily make sense of your data files?